Phase-aware symbolic quantum-gate synthesis with operator-Schmidt certificates

Anonymous referee draft

7 November 2026

Abstract

We study exact symbolic synthesis of small quantum operators into supplied native gate alphabets, distinguishing literal matrix equality from equality up to a recorded scalar. Operator-Schmidt ranks guide factorization and interaction search; native analytical charts and exact local rewrites supply candidate circuits; separate certificates establish the scope of optimality. The released reference grid contains 327 catalog entries and 32 presets, or 10,464 target–basis cells under two phase policies. Each phase mode has an exact construction or an obstruction to representing the requested target or full parameter family. We give a four-CNOT circuit for controlled fermionic SWAP and a matching lower bound for arbitrary Ising-class interactions with unrestricted local gates and no ancillas. Explicit, branch-complete symbolic charts attain the known two-B and three-SQiSW uniform bounds. A four-qubit parameter-reuse experiment reduces 36 CNOTs to 18 relative to the tested compile-once Qiskit route. The accompanying atlases and archives retain expressions, scalar conventions, proof scopes, and implementation provenance. The finite grid is classified completely; unrestricted local-cost optimality and superiority to every compiler or published circuit are not established.

1 Introduction

Parameterized quantum subroutines must often be expressed using a different entangling primitive while preserving their parameters and their behavior inside a larger computation. Three distinct tasks arise: obtaining a valid identity, finding an economical circuit in the selected alphabet, and proving that no cheaper circuit exists. A small numerical error in a bound instance does not certify a free-parameter identity. Conversely, a symbolic identity does not certify a minimum. Global phase adds a further resource distinction: a scalar that is harmless for an isolated operator becomes a relative phase when that operator is controlled.

This work develops an auditable synthesis and reference workflow for these tasks. It combines exact operator-Schmidt analysis with native constructions and local circuit transformations. The output includes a circuit, its phase convention, a parameter-sensitive score when evaluated, and explicitly scoped lower bounds. The target is a reusable library of exact small-block identities, rather than a claim to solve arbitrary continuous circuit optimization.

Results and assumptions. The reference grid has 327 catalog entries and 32 mathematical presets. All 20,928 phase modes are classified as a verified symbolic construction or a proved obstruction. The latter can exclude representation of an entire continuous family without excluding every special parameter value. The grid contains aliases and fixed-width register fixtures; it is not a collection of 327 independent gate families or 32 calibrated processors. We preserve the supplied

alphabets, use no ancillas, allow arbitrary input states and all-to-all coupling, and fix input and output wire order. No physical global phase gate is permitted. Our principal concrete results are:

- a literal four-CNOT controlled-fermionic-SWAP construction with a matching four-Ising lower bound, including arbitrary interaction angles;
- explicit symbolic local frames and degeneracy branches implementing the known two-B and three-SQiSW uniform two-qubit bounds;
- exact-phase Clifford shortest-path certificates and separately scoped native entangler and secondary-score certificates;
- independent Qiskit/Cirq comparisons, bounded component ablations, an application with reusable parameters, and editable circuit atlases.

We distinguish improvements over a particular SDK output, attainment of a known analytical bound, and a proved optimum. None alone establishes publication priority. In particular, the four-Ising theorem is presented as an explicit result to scrutinize, without a claim that no equivalent result appears elsewhere.

2 Related work and positioning

Two-qubit local invariants and Cartan decompositions provide established tools for recognizing and constructing economical circuits [1, 2]. The B gate admits a universal two-pulse construction [3]. Huang et al. characterize SQiSW instruction sets and give the three-pulse universal bound and the two-pulse region [4]. Those counts are prior results. Our implementation contribution is a collection of explicit symbolic outer frames, scalar handling, degeneracy branches, and replayable certificates integrated into a larger catalog workflow.

Symbolic circuit optimization also predates this work. Quartz generates and verifies gate-set-dependent transformations and applies cost-based search [5]. QUESO synthesizes symbolic rewrite rules using polynomial identity filtering and applies them through beam search [6]. These systems address a substantial part of the motivation for reusable parameterized transformations. We do not claim that symbolic motif discovery, heterogeneous instruction sets, or compiler-generated rewrite rules are new concepts. Our emphasis is the combination of exact phase policies, OSR-guided decomposition, native analytical charts, scoped minimum certificates, and a directly inspectable reference atlas.

UniversalQCompiler is an analytical Mathematica synthesis package, with unitary, isometry, and channel compilation and LaTeX circuit export [7]. It is relevant analytical prior art. No runtime or circuit-quality experiment against it is reported here because a licensed kernel was unavailable. Quartz and QUESO were not executed either; we claim no measured dominance over those tools. Our SDK experiments are restricted to the versions and routes recorded below.

Qiskit already represents symbolic parameters and supports compile-once parameter binding [8]. The use case here is reducing the symbolic native circuit before repeated binding, not introducing parameter reuse. Earlier OSR-based partition resynthesis [11] concerns numerical block optimization and globally optimal selection over enumerated partition candidates. Its numerical sample is not reused as evidence for the symbolic identities in this work.

3 Problem, resources, and phase

Let $U(\vec{\theta})$ be an exact $2^n \times 2^n$ unitary matrix over real symbolic parameters and let $\mathcal{B}$ be a supplied gate alphabet. An output word C consists only of gates in $\mathcal{B}$, on distinct valid wires, with exact expressions and no floating constants. Literal mode requires $U = C$. Projective mode requires

$$U = e^{i\gamma(\vec{\theta})}C, \qquad |e^{i\gamma}| = 1, \tag{1}$$

with the scalar stored as metadata at zero physical cost. A literal circuit is preferred when the two costs agree. If the projective word is strictly cheaper, both are retained; unresolved score comparisons also retain both. Literal scalar absorption, when possible, uses allowed local gates.

We order resources lexicographically as (E, S, L): number of native two-wire operations, parameter-sensitive score, then number of one-wire operations. For an operation g acting on k_g wires with explicit parameters p_j, objective version 2 uses

$$S(C) = \sum_{g \in C} \left[2^{k_g} + \frac{1}{5} \sum_j \left(\log_2 \max\{1, |\operatorname{den}(\operatorname{simplify} p_j)|\} + \frac{1}{100} \right) \right]. \tag{2}$$

A fixed named S or T has no explicit angle surcharge; a U3 has three. This is a symbolic-description preference, not a calibrated pulse model. The normal form chosen by the simplifier is part of the scoring contract. Denominator complexity is not invariant under all mathematical expression representations. Candidate ordering uses numeric evaluations of recorded exact score expressions; exact identity acceptance is independent of that ordering.

The nonnegative clamp is necessary. In the historical unclamped objective, put $d = \log(2049/2048)$ and $a = 1/d$. The exact identity word $U_3(-a, 0, 0)U_3(a, 0, 0)$ has negative score $1003/250 + (2/5)\log_2 d$. Repeating it makes unrestricted minimization unbounded below. The revised objective removes this obstruction; an exact migration audit preserved all scores for 1,390 distinct retained angle expressions. Nonnegativity gives finite word-length bounds below an incumbent score, but does not by itself solve continuous angle minimization.

Some shared-expression charts are too large for full score evaluation within the recorded budget. Their count E remains exact, and a strict improvement in E decides the lexicographic order. At equal E, an unknown score is incomparable. We never score a temporary alias as if it were the underlying angle, set an unknown score to zero, or use L to break an unresolved secondary comparison.

4 OSR-guided symbolic synthesis

4.1 Exact cuts, factors, and lower bounds

For a bipartition $A|B$, realign $U_{ab,a'b'}$ into a matrix with row index (a, a') and column index (b, b'). Its rank is the operator-Schmidt rank $r_{A|B}(U)$. The implementation works over exact exponential rational functions; it does not threshold numerical singular values. Rank-one realignment gives a tensor factorization. All nontrivial bipartitions are considered once, choosing the side containing local wire zero as the canonical representative. This cut enumeration is exponential in width; we make no polynomial scaling claim for unbounded n.

The basic inequalities are

$$r_{A|B}(UV) \le r_{A|B}(U) r_{A|B}(V), \qquad r_{A|B}(L_A \otimes L_B) = 1. \tag{3}$$

If every native interaction crossing a cut has rank at most ρ, at least $\lceil \log_\rho r_{A|B}(U) \rceil$ crossings are necessary. A nonzero exact minor records the rank witness and, for a family, the condition under which it is nonzero. An exact hard specialization can prove a necessary uniform budget for the entire family. It cannot prove that every parameter value needs that budget.

4.2 Candidate construction and acceptance

The constructor first checks resource obstructions and elementary/native matches, then exposes independent factors or controlled structure where possible. Residual search peels candidate interactions, evaluates the OSR profile of the residual, and explores small interaction patterns. Catalog identities, controlled blocks, exchange structures, Clifford frames, and analytical native charts supply complementary candidates. Local gates are fused, lowered to the exact supplied alphabet, and simplified using independently proved identities. The final selection compares only accepted candidates under the declared resource order.

Search is not an exhaustive enumeration of all continuous circuits. Numerical optimization may propose a fixed circuit or an angle relation; it is retained only after exact lifting and an exact identity proof. Genetic programming is not required for the released certificates. Catalog-specific dispatch and opaque-matrix synthesis are distinguished in the experiments, since success on named templates alone would be weak evidence of generalization.

Acceptance uses exact matrix identities, reductions to polynomial/trigonometric identities, or a compositional certificate consisting of a proved parent and proved replacements. Algebraic roots and branch choices carry explicit conditions. A failure to decide an identity is unresolved, not false or proved impossible. Time limits are reported independently from feasibility.

4.3 Native charts and shared expressions

We use $\mathcal{A}(x, y, z) = \exp[i(xXX + yYY + zZZ)]$. The B chart constructs two $B = \mathcal{A}(\pi/4, \pi/8, 0)$ pulses with explicit U3 frames, folds, corner branches, and scalar absorption. Its full-family matrix identity reduces to exact Pauli coefficients and normalization polynomials.

For SQiSW $= \mathcal{A}(\pi/8, \pi/8, 0)$, the two-pulse middle chart lies in the known region $x \geq y + |z|$ of the Weyl chamber [4]. The constructor recovers its outer frames symbolically using the real and imaginary Pauli tensors of the spin-flip matrix. Known symbolic eigenvalues yield spectral projectors of a real 3×3 tensor; explicit pivot choices handle degenerate subspaces. Periodic-angle folding and a third pulse cover the remaining chamber. Euler and scalar absorption charts complete native lowering. The detailed equations and branch certificates are supplied in the separate SQiSW proof supplement.

Five broad families—Canonical, fSim, PhasedFSim, SWAPPow, and TK2—now use three SQiSW interactions in each of two presets and both phase modes, replacing six-interaction incumbents. Each family contains a SWAP-class specialization outside the two-pulse region, so three is the minimum *uniform* budget. The exact circuits use nine local gates and the projective circuits eight; these are counts, not minimum-local-cost claims. The approximately 600 scalar definitions per chart are stored as a scoped acyclic expression program. Expanding repeated subexpressions can produce hundreds of millions of expression-tree occurrences. A diagram label is therefore linked to its exact program and glossary. The retained 20 phase results have unresolved secondary scores.

5 A four-Ising optimum for controlled fermionic SWAP

Let F act on two qubits by

$$F|00\rangle = |00\rangle, \quad F|01\rangle = |10\rangle, \quad F|10\rangle = |01\rangle, \quad F|11\rangle = -|11\rangle. \tag{4}$$

Define $U_F = |0\rangle\langle 0|_c \otimes \mathbb{I} + |1\rangle\langle 1|_c \otimes F$. This is controlled fermionic SWAP, not ordinary Fredkin. The extra sign changes its synthesis problem. In our little-endian indexing the control is wire zero, with basis states 3 and 5 interchanged and basis state 7 negated.

Theorem 1 *The operator U_F requires exactly four CNOTs with arbitrary one-qubit unitaries, no ancillas, and all-to-all connectivity. More generally, any controlled two-qubit unitary* $\mathrm{diag}(\mathbb{I}, V)$ *with* $\mathrm{OSR}(V) = 4$ *requires at least four Ising-class two-qubit interactions, even allowing different interaction angles, arbitrary local unitaries, and an arbitrary scalar.*

Here an Ising-class interaction is any unitary locally equivalent to $\exp(itZ \otimes Z)$, including CNOT, CZ, ECR and tunable XX, ZZ, or ZX interactions at entangling angles. The lower bound concerns that relaxed class; a fixed basis needs its own matching upper construction.

5.1 Upper construction

The following word, read in time order, is literally U_F:

1	$R_y(\pi/2)_1$	8	$\mathrm{CX}_{0,2}$
2	$R_z(\pi)_2$	9	$R_y(-\pi/4)_2$
3	$\mathrm{CX}_{1,2}$	10	$\mathrm{CX}_{1,2}$
4	$R_y(\pi/4)_1$	11	$R_y(\pi/2)_1$
5	$R_y(-3\pi/4)_2$	12	$R_z(\pi)_0$
6	$\mathrm{CX}_{0,1}$	13	$R_z(\pi)_1$
7	$R_y(-\pi/4)_1$	14	$R_x(-\pi)_2$

Exact matrix multiplication proves the identity with no GP gate or unrecorded scalar. The native lowering pipeline also produces a CX/U3 word with $E = 4, L = 10$. The latter is a local-count upper bound. The raw archive preserves the literal word and independent rank witnesses.

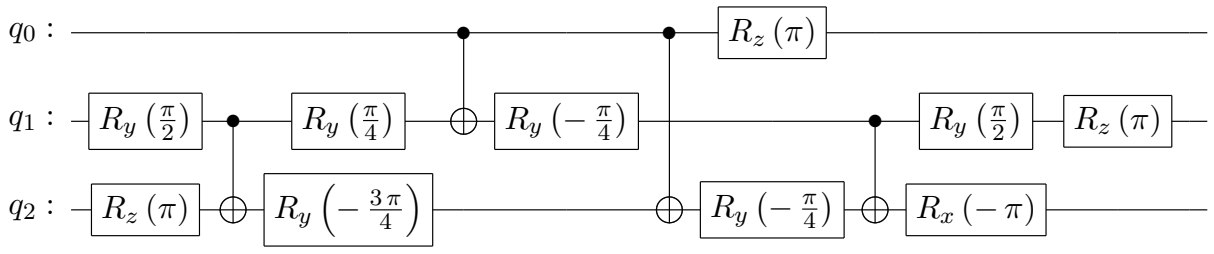


Figure 1: The literal four-CNOT construction. Wire zero is the control; the final local gates implement the exact phase convention.

5.2 Control-space rigidity

Lemma 1 *If invertible one-qubit matrices P, Q map the full diagonal matrix space*

$$\mathcal{D} = \mathrm{span}\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$$

onto itself by $D \mapsto PDQ$, then P, Q are monomial with the corresponding common permutation. Thus locally equivalent controlled operators with control OSR two have corresponding conditional target blocks, up to phases, exchange of the two branches, and common target-local input/output frames.

Proof. The only rank-one lines in $\mathcal{D}$ are its two coordinate projectors: $\det \mathrm{diag}(a, b) = ab$. An invertible left/right map preserves rank and must permute these two lines. Its column and row factors therefore have the stated form. Control Schmidt spaces of rank-two controlled operators are full copies of $\mathcal{D}$, proving the second claim. In particular, a locally equivalent realization of $\mathrm{diag}(\mathbb{I}, V)$ has a local conditional branch. Its relative target block has the same OSR as V, also after branch inversion. *QED*

5.3 Exclusion of three interactions

Proof of the lower bound in Theorem 1. Call the targets a, b. The singleton cuts have ranks $2, 4, 4$ for c, a, b: the rank-four assertion follows from the V block, since a target cut has maximum rank four. Each crossing Ising interaction has rank at most two. Consequently a and b each have degree at least two and c at least one in an interaction graph. Two interactions cannot satisfy these demands. With three, the only edge multisets are a triangle ca, cb, ab, or two ab edges and one control edge. There are six time orders in each class, twelve in total.

First consider one control crossing and two internal target interactions. Absorb local frames into adjacent gates and put the single crossing in controlled form. Its target branches are W_+W_- and W_+QW_-, where Q is a one-wire unitary and $W_\pm$ contain the target operations after/before the crossing. By Lemma 1, choose a local branch as the first one (inverting Q if the branches are exchanged). If both internal interactions lie on one side, the corresponding product must be local, so the relative branch is local. If one lies on each side, the relative branch is $W_-^\dagger QW_-$ with W_- locally Ising. Conjugation of a one-wire operator by e^{itZZ} has support on the other wire in $\mathrm{span}\{\mathbb{I}, Z\}$, hence OSR at most two. Local frames do not change this bound. In neither case can the relative block have rank four.

Now consider the triangle. Put each control interaction in controlled form and combine all operations on the control between them into a one-qubit unitary N. Local frames at the endpoints are harmless by Lemma 1. A 2×2 unitary is either monomial or has all four entries nonzero. If N is monomial, the circuit remains controlled after a possible control permutation. Each conditional target branch contains the sole internal ab interaction, with only local factors around it. That interaction must be entangling; otherwise the singleton rank demands already fail. Neither branch is local, a contradiction.

If N has all entries nonzero, factor the target operation K between the crossings to one side. The four control blocks then span $\{\mathbb{I}, B, A, AB\}$, where B is a nontrivial one-wire unitary on one target and A is the other controlled target action conjugated through K. Their span dimension equals the control OSR, namely two. If the internal interaction is outside the crossings, K is local, and A, B act nontrivially on different wires. The four tensor products are independent, contradicting rank two. If the internal interaction lies between them, independence of $\mathbb{I}, B$ forces $A \in \mathrm{span}\{\mathbb{I}, B\}$. The transformed controlled gates then act only on c, b. Apart from local factors the whole circuit

is $(\mathbb{I}_c \otimes K_{ab})W_{cb}$. Its $a|cb$ rank is at most $\mathrm{OSR}(K) = 2$, again a contradiction. These cases exclude every three-interaction word. The explicit four-CNOT identity attains the bound for U_F. *QED*

The supplied checker verifies the exact matrix identity, cut minors, three-edge enumeration, Ising conjugation identities, and native-class identifications. The argument above is an analytical proof with checked algebraic components, not a proof-assistant formalization or a conclusion from failed numerical optimization.

6 Other minimum and obstruction certificates

6.1 Two-qubit nonlocal invariants

Write $\Gamma = U(YY)U^T(YY)$ and $\Delta = \det U$. The normalized invariants

$$g_1 = \frac{(\mathrm{tr}\,\Gamma)^2}{16\Delta}, \qquad g_2 = \frac{(\mathrm{tr}\,\Gamma)^2 - \mathrm{tr}(\Gamma^2)}{4\Delta} \tag{5}$$

are unchanged by local gates and scalar phase [1]. Their one-native-pulse values exclude that pulse class when unequal. Additional two-pulse constraints for Ising, iSWAP, SQiSW and square-root-SWAP supply lower bounds. For example, two SQiSW pulses require

$$W = g_2 + 1 \geq 0, \qquad W^2 \geq 8(|g_1| + \Re g_1). \tag{6}$$

An exact violation excludes two pulses; a matching three-pulse chart gives an optimum. The proof supplement derives the polynomial identities and separately excludes zero/one-pulse cases where needed. We do not transfer a fixed primitive's lower bound to an arbitrary tunable gate without proving the larger class condition.

6.2 Clifford phases and secondary floors

The atlas includes 24 projective single-qubit Cliffords and their eight scalar phases, giving 192 literal matrices. Finite-alphabet graph searches store exact matrices and shortest-path certificates. In the $\{H, S, T, T^\dagger\}$ alphabet, a radius-seven exact search reaches 628 matrices, including non-Clifford intermediates. Meet-in-the-middle lookup proves all 192 word minima with maximum length fourteen: every shorter word splits into two radius-seven halves. Eighth-root phase completeness follows from the coefficient field $\mathbb{Q}(\sqrt{2}, i)$, excluding missing sixteenth-root corrections. Continuous alphabets are outside this finite proof; their atlas circuits are not all claimed globally local-optimal.

Under Eq. (2), every local costs at least two. Nonzero commutators with embedded X or Z witness wires on which a zero-entangler target acts nontrivially. Combining support bounds with mandatory parameter surcharges certifies a secondary minimum when a construction meets that floor. These give 372 literal and 1,144 projective global lexicographic certificates in the preserved floor audit, with overlaps with other proofs. We report these separately from the entangler-only coverage.

6.3 Feasibility obstructions

A finite discrete alphabet cannot represent a nonconstant continuous family by a single unrestricted exact-family construction. Determinants of allowed circuits lie in a generated subgroup; violation of its necessary root condition excludes literal equality. Clifford normalization and the field $\mathbb{Q}(\sqrt{2}, i)$ give further necessary conditions for Clifford and Clifford+T alphabets. Projective tests remove arbitrary scalar where appropriate. The archive records a witness and scope. A failed search

is never converted into an impossibility record, and a generic-family obstruction permits exactly representable specializations.

7 Evaluation

7.1 Frozen grid and certificate scope

Classification / selected certificate	Literal	Projective
Symbolic construction	7,702	10,196
Target/family obstruction	2,762	268
Selected fixed-target entangler minima	712	959
Selected uniform-family budget minima	720	965

The selected-certificate rows take the union of the independent native/exchange audits, the new SQiSW family witnesses, and the four-Ising certificates. They exclude other OSR, Clifford and support-floor certificates and are not additive with those counts. The grid comprises 243 fixed entries, 64 free-real-parameter families, and 20 register fixtures, each crossed with 32 presets. The archive records per-mode source-run indices because successive refinements use different implementations. A complete classification means that the requested catalog/family contract is decided; it does not establish a complete classification of every exceptional parameter specialization. Some early runs recorded a Git HEAD preceding uncommitted source edits. The verifier resolves their recorded source-content hashes from Git history and lists the differing content revisions explicitly; the original recorded HEAD is not silently relabeled as containing those edits.

The new release replaces the twenty SQiSW frontier results and the fFredkin rows while preserving earlier raw attempts and source snapshots. Old independent proof reports remain tied to their immutable input; their statements transfer only for the same target, basis, phase policy, and matching count. The release verification report checks these conditions. Lower-bound timeouts are recorded in the older proof audit even though no retained synthesis mode is a timeout. Thus "no unresolved synthesis cells" is narrower than "all research questions are solved."

7.2 Independent fixed-target baselines

Qiskit is run on two independently constructed inputs: an exact target matrix converted numerically, and a library source word. Neither input is our optimized circuit. We use CX/U and CZ/U bases, optimization level three, approximation degree one, seeds 7 and 219, arbitrary inputs and no ancillas. Output permutations are checked; if a virtual permutation is introduced, the wrapper reruns level one and records the actual level. Full numerical matrix replay includes Qiskit's scalar. Cirq uses its full-CZ target gateset and independent library sources, with explicit little-endian conversion for fFredkin. Cirq replay is projective. Numerical SDK circuits are comparison upper bounds, not symbolic proofs.

Target	Tool	Input	Basis	E	L
SYC	cirq	library_source	cz	3	8
SYC	qiskit	library_source	cx	3	8
SYC	qiskit	library_source	cz	3	8
SYC	qiskit	matrix	cx	3	8
SYC	qiskit	matrix	cz	3	8
fFredkin	cirq	library_source	cz	17	43
fFredkin	qiskit	library_source	cx	13	17
fFredkin	qiskit	library_source	cz	13	19
fFredkin	qiskit	matrix	cx	7	17
fFredkin	qiskit	matrix	cz	7	16
SYC	This work	exact identity	CX/U	3	3
fFredkin	This work	exact identity	CX/U	4	10

Both Qiskit seeds give the same displayed counts. Cirq uses projective replay; Qiskit replay includes its scalar. Raw operations, actual optimization levels and residual errors are archived. The earlier broad comparison contains 2,673 fixed-target/basis pairs on supported Qiskit routes, with 11 lower entangler counts and 2662 ties in this release. Some of its inputs were source-fed constructions, so those ties are not independent discoveries. The focused experiment above addresses this limitation for its two targets. A tested route is not the best result obtainable with every SDK pass manager.

At equal entangler count, the broader local-count audit has 553 reductions, 2108 ties and 1 higher local counts. In particular, C3X/ZZ/U3 retains 28 locals with score 115.594 while an available 24-local alternative scores 116.37. This is a tradeoff between two verified witnesses under the declared secondary objective, not a proof that no unknown circuit improves both. We make no blanket local-count dominance claim.

7.3 A scoped Sycamore comparison

For SYC = fSim$(\pi/2, \pi/6)$ the projective Rz/CX word has three locals and three CNOTs. Its tracked factor in Eq. (1) is $e^{-7\pi i/24}$. Figure 3 of AbuGhanem and Eleuch [9] displays three CNOTs and 27 local boxes (15 Rz and 12 SX). The 27-to-3 comparison is against that specific displayed circuit, under projective equivalence. It is not a claim to improve every published Sycamore decomposition. The independent SDK experiment yields three entanglers and eight locals for the tested routes. Abstract local boxes are not microwave layers: virtual Z rotations can have zero duration [10]. No latency, fidelity, or error reduction is inferred from these counts.

7.4 Bounded opaque-family ablations

Six structurally perturbed two-qubit families, absent as named catalog entries, are tested in SQiSW/U3 and B/U3 presets. They include X/Y-controlled rotations, unequal controlled SU(2) branches, phase-gauged number-conserving blocks, an affine Cartan family, and unequal commuting exchanges. The full and chart-disabled methods receive opaque exact matrices; no template name is supplied. A third route receives an independent exact source word and only lowers that word. Each method/case/preset has a 20-second budget. The OSR residual-search budget is 0.2 seconds; identity certification and native lowering share the outer deadline. All source words are independently proved equal to their matrices.

Method	Verified	Unresolved	Deadline	Errors
Full matrix synthesis	11	0	1	0
Native charts disabled	3	9	0	0
Independent source lowering	12	0	0	0

The full method verifies 11 of 12 case–preset pairs within the budget; the gauged affine Cartan/B case exhausts its 20 seconds. Source lowering verifies all 12, generally using more entanglers. This bounded failure is retained. Disabling native charts leaves the other search machinery intact,

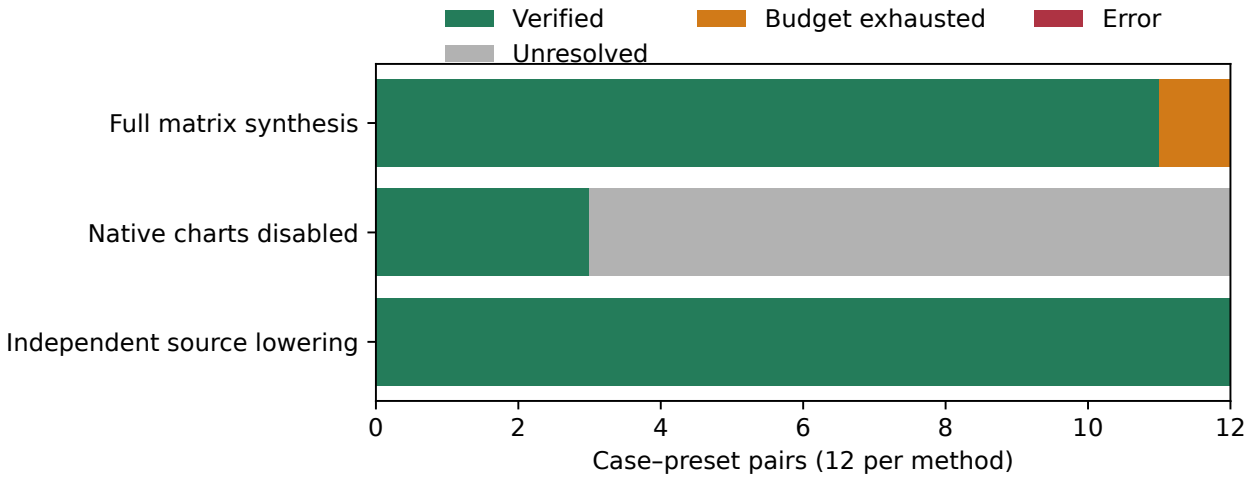


Figure 2: Bounded component ablation. Verified counts, unresolved searches, and exhausted budgets are kept distinct. Resource summaries include only verified outputs; missing results are not zero-cost circuits.

including OSR and local lowering. This is a native-chart ablation, not a clean "OSR off" experiment. The examples are deliberately related structural perturbations, not a blind random distribution or an independent benchmark of all possible symbolic operators. Their result supports the role of the charts in this bounded workload, not universality of the automatic search.

7.5 Reusable parameters in a four-qubit circuit

We compose six fSim blocks on $(0,1),(2,3),(1,2)$ repeated twice, with twelve independent real parameters. The baseline source uses the exact identity $\mathrm{fSim}(\theta,\phi) = R_{xx}(\theta)R_{yy}(\theta)\mathrm{CP}(-\phi)$. Both methods use the same compile-once Qiskit settings. Our method first replaces each block by its literal three-CNOT symbolic chart. No known input state, output permutation, or approximate synthesis is used.

Route	E	L	Depth	SDK compile (s)	1,000 binds (s)
SDK source	36	66	55	0.0508	0.1143
Symbolic source	18	42	31	0.0071	0.0521

Cold symbolic chart synthesis takes 5.628 seconds in this run, in addition to the symbolic route's SDK compile time. Consequently the cold symbolic workflow is slower here despite its smaller circuit; amortized timing depends on reuse. The largest of the ten full-matrix replay errors is 1.6×10^{-14}. We time cold symbolic chart construction, SDK compilation, and 1,000 parameter assignments separately. Ten sampled assignments also receive full 16-by-16 matrix replay, and

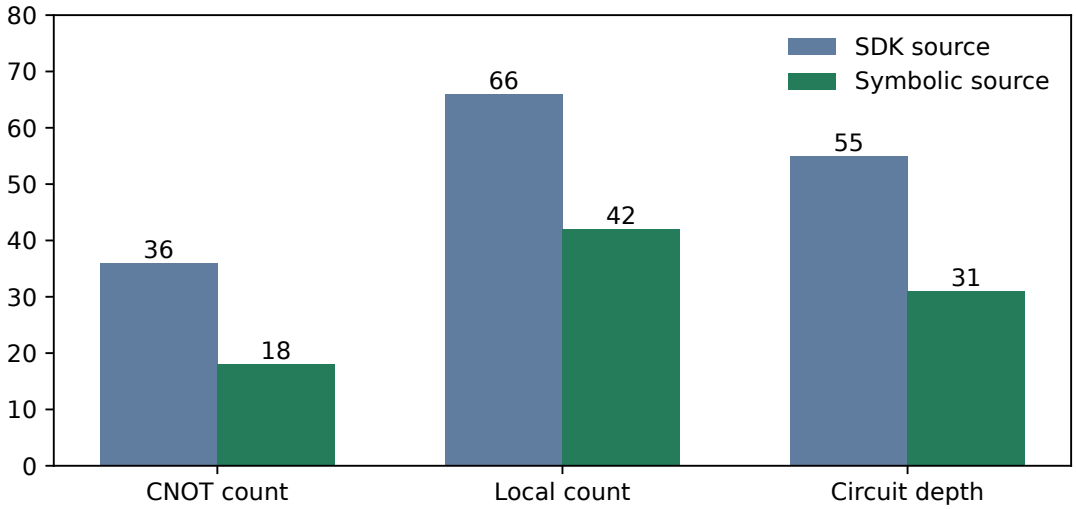


Figure 3: Counts after the same SDK optimization of the baseline source and the symbolically lowered source. This is one synthetic parameter-reuse example; hardware runtime and fidelity were not measured.

are independently numerically recompiled as a separate diagnostic. The family identity is proved compositionally before those samples; the samples are regression checks. This is a single run of a small synthetic workload, so timings are descriptive and not a robust end-to-end speedup estimate.

7.6 Controlled embedding audit

For the three-Rz Sycamore circuit C, the exact identity is $\mathrm{SYC} = e^{-7\pi i/24}C$. Controlling C alone fails literal equality to controlled Sycamore. Adding $P_c(-7\pi/24) = \mathrm{diag}(1, e^{-7\pi i/24})$ on the new control restores the identity exactly. The audit stores a nonzero matrix-entry witness for the uncorrected substitution and proves the corrected identity. It demonstrates why the phase contract matters; it is not evidence that SDKs with existing global-phase tracking handle control incorrectly.

8 Artifacts, limitations, and conclusions

The repository is https://github.com/rakytap/sequential-quantum-gate-decomposer. This referee package includes the source-pinned result archives, build commands, environment versions, exact proof payloads, comparison circuits, and editable LaTeX. It also includes a catalog atlas with 32 preset cells per gate page, exact-expression glossaries, an alphabetical index, and an all-phase Clifford atlas with 24 circuits per page. Aliases are labeled, both phase policies are retained when needed, and each shared-expression label resolves to its full definition. The release manifest records hashes of the frozen inputs and generated artifacts. No permanent archival DOI has yet been assigned; a tagged archival deposit is a publication-stage task rather than an existing reproducibility claim.

The most important limits are mathematical and experimental. General continuous local-cost optimality is open. A lower bound can match a uniform family budget without matching every specialization. Twenty shared-expression secondary scores remain unevaluated. Finite alphabet proofs do not imply continuous alphabet minima. Preset counts do not include calibration, scheduling, connectivity overhead, or restricted hardware angle domains beyond the specified mathematical

model. The focused independent comparison and six-family ablation do not establish dominance over all pass managers, UniversalQCompiler, Quartz, QUESO, or all literature. The lower-bound argument in Theorem 1 warrants independent mathematical scrutiny; its component checker does not replace that scrutiny.

Within these limits, the workflow produces a complete classified reference grid, a four-Ising optimum for controlled fermionic SWAP, explicit native symbolic charts attaining known bounds, and a reproducible parameterized application improvement. Keeping construction, score, phase, and proof scope separate makes the resulting circuit library useful even where an unrestricted optimum remains unproved.

Author contributions and use of AI. Author names, affiliations, and the allocation of human contributions are withheld in this anonymous referee draft and must be supplied and approved by the authors before submission. An OpenAI Codex coding assistant was used for software implementation, experiment orchestration, calculations, bibliographic research, and manuscript drafting. The accompanying exact checks and raw records are provided for independent review; they do not assign authorship or responsibility to an AI system. All named authors must verify and take responsibility for the submitted work.

References

[1] Y. Makhlin, Nonlocal properties of two-qubit gates and mixed states and optimization of quantum computations, Quantum Information Processing 1, 243–252 (2002), arXiv:quant-ph/0002045.

[2] V. V. Shende, S. S. Bullock, and I. L. Markov, Recognizing small-circuit structure in two-qubit operators and timing Hamiltonians to compute controlled-NOT gates, Physical Review A 70, 012310 (2004), doi:10.1103/PhysRevA.70.012310.

[3] J. Zhang, J. Vala, S. Sastry, and K. B. Whaley, Minimum construction of two-qubit quantum operations, Physical Review Letters 93, 020502 (2004), arXiv:quant-ph/0312193.

[4] C. Huang et al., Quantum instruction set design for performance, Physical Review Letters 130, 070601 (2023), arXiv:2105.06074.

[5] M. Xu, Z. Li, O. Padon, S. Lin, J. Pointing, A. Hirth, H. Ma, J. Palsberg, A. Aiken, U. A. Acar, and Z. Jia, Quartz: Superoptimization of quantum circuits, Proceedings of PLDI (2022), doi:10.1145/3519939.3523433.

[6] A. Xu, A. Molavi, L. Pick, S. Tannu, and A. Albarghouthi, Synthesizing quantum-circuit optimizers, Proceedings of the ACM on Programming Languages 7, PLDI, Article 140 (2023), doi:10.1145/3591254.

[7] R. Iten, O. Reardon-Smith, E. Malvetti, L. Mondada, G. Pauvert, E. Redmond, R. S. Kohli, and R. Colbeck, Introduction to UniversalQCompiler, arXiv:1904.01072 (2019; revised 2021).

[8] IBM Quantum, Qiskit Parameter documentation, `https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.circuit.Parameter`, accessed 7 November 2026; executed versions are in the experiment manifest.

[9] M. AbuGhanem and H. Eleuch, Full quantum tomography study of Google's Sycamore gate on IBM's quantum computers, EPJ Quantum Technology 11, 36 (2024), doi:10.1140/epjqt/s40507-024-00248-8.

[10] D. C. McKay, C. J. Wood, S. Sheldon, J. M. Chow, and J. M. Gambetta, Efficient Z gates for quantum computing, Physical Review A 96, 022330 (2017), arXiv:1612.00858.

[11] G. Morse, M. Al Farizi, J. Nádori, L. Hajas, T. Kozsik, and P. Rakyta, Entanglement-aware quantum circuit optimization through globally optimal partitioning, arXiv:2611.03845 (2026).

A Preset definitions and interpretation

Preset	Entangler	One-qubit alphabet
cx_u	CX	U3
cx_rz_ry	CX	Rz, Ry
cx_h_phase	CX	H, P
ibm_cx	CX	Rz, Sx, X
ibm_ecr	ECR	Rz, Sx, X
ibm_cz	CZ	Rz, Sx, X
cz_u	CZ	U3
cz_rz_rx	CZ	Rz, Rx
iswap_u	iSWAP	U3
sqrt_iswap_u	SiSWAP	U3
sqrt_swap_u	SSWAP	U3
xx90_u	XX90	U3
xx_rz_ry	Rxx	Rz, Ry
zz_u	Rzz	U3
zx_u	Rzx	U3
google_cz	CZ	PhasedXZ
google_sqrt_iswap	SiSWAP	PhasedXZ
google_syc	SYC	PhasedXZ
google_fsim	fSim	PhasedXZ
berkeley_b	B	U3
ionq_aria	IonQMS	GPi, GPi2
ionq_aria_full	IonQMSMax	GPi, GPi2
ionq_forte	IonQZZ	GPi, GPi2
quantinuum_zzmax	ZZMax	PhasedX, Rz
quantinuum_zzphase	ZZPhase	PhasedX, Rz
tket_tk2	TK2	PhasedX, Rz
rigetti_cz	CZ	Rz, Sx, X
aqt_rxx	Rxx	R, Rz
iqm_cz	CZ	R
clifford_t_cx	CX	H, S, T, Tdg
clifford_t_cz	CZ	H, S, T, Tdg
clifford_only	CX	H, S

Each primitive name denotes the exact catalog convention. "U" denotes the U3 local chart; XX/ZZ/ZX angles use the catalog's half-angle convention where specified. TK2 is counted as one supplied macro primitive, not silently equated with a single calibrated pulse. Fixed discrete alphabets are preserved even when they obstruct the requested continuous family. Single-entangler-type presets are distinct from requiring only one entangler in a circuit.

B Experiment detail and reproduction

Software	Version
Python	3.9.25
numpy	1.26.4
scipy	1.13.1
sympy	1.14.0
qiskit	2.2.3
cirq-core	1.3.0

The experiment source revision is `b3ba5e26cb75`. Four fork workers run the bounded jobs with `OPENBLAS_NUM_THREADS=1`. Timings include concurrent-worker load and are single-run observations. From the pinned repository revision, the bounded experiments are reproduced with:

```
OPENBLAS_NUM_THREADS=1 python3 \
 examples/decomposition/symbolic_manuscript_experiments.py \
 --output /tmp/manuscript-experiments.json.gz \
 --workers 4 --seconds 20 --bindings 1000
```

Tables and plots are regenerated by `examples/decomposition/build_symbolic_manuscript.py`; its default inputs are the frozen archives. Timings depend on the machine; resource counts and exact identities are the portable claims. The release verification script checks source snapshots, changed identities, partial score policy and artifact integrity. The full symbolic test suite is `python3 -m pytest tests/symbolic -q`.

Case	Preset	Method	E	Seconds	Status
Affine Cartan	B	Full	–	20.000	budget_exhausted
Affine Cartan	B	Source	6	5.190	verified_symbolic
Affine Cartan	B	No chart	–	0.467	unresolved
Affine Cartan	SQiSW	Full	3	5.431	verified_symbolic
Affine Cartan	SQiSW	Source	6	4.103	verified_symbolic
Affine Cartan	SQiSW	No chart	–	0.828	unresolved
Gauged exchange	B	Full	2	12.699	verified_symbolic
Gauged exchange	B	Source	4	3.141	verified_symbolic
Gauged exchange	B	No chart	2	3.402	verified_symbolic
Gauged exchange	SQiSW	Full	2	10.588	verified_symbolic
Gauged exchange	SQiSW	Source	2	0.824	verified_symbolic
Gauged exchange	SQiSW	No chart	2	1.784	verified_symbolic
Unequal exchange	B	Full	2	20.000	verified_symbolic
Unequal exchange	B	Source	8	7.634	verified_symbolic
Unequal exchange	B	No chart	2	4.369	verified_symbolic
Unequal exchange	SQiSW	Full	3	1.849	verified_symbolic
Unequal exchange	SQiSW	Source	8	4.450	verified_symbolic
Unequal exchange	SQiSW	No chart	–	0.518	unresolved
Unequal branches	B	Full	2	7.697	verified_symbolic
Unequal branches	B	Source	8	10.506	verified_symbolic
Unequal branches	B	No chart	–	1.057	unresolved
Unequal branches	SQiSW	Full	2	11.968	verified_symbolic
Unequal branches	SQiSW	Source	8	5.429	verified_symbolic
Unequal branches	SQiSW	No chart	–	1.035	unresolved
X-control	B	Full	2	14.652	verified_symbolic
X-control	B	Source	4	5.001	verified_symbolic
X-control	B	No chart	–	0.400	unresolved
X-control	SQiSW	Full	2	16.135	verified_symbolic
X-control	SQiSW	Source	4	2.973	verified_symbolic
X-control	SQiSW	No chart	–	0.436	unresolved
Y-control	B	Full	2	14.181	verified_symbolic
Y-control	B	Source	4	5.000	verified_symbolic
Y-control	B	No chart	–	0.624	unresolved
Y-control	SQiSW	Full	2	16.039	verified_symbolic
Y-control	SQiSW	Source	4	2.568	verified_symbolic
Y-control	SQiSW	No chart	–	0.577	unresolved